Deception+ v1

A proof-of-concept model of delivery/release-based deception

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This document exists to answer one question honestly: given only what's real and obtainable, is a delivery/release-based deception model actually useful? The verdict below is not decided in advance - it's read directly off the validation numbers in Section 4.

1. The Verdict

Useful, with real caveats. The feature set is thin, but it's measuring something stable and predictive beyond Stuff+.

Split-half reliability (100+ pitch pitchers): **0.841**. Outcomes with a meaningful partial correlation beyond Stuff+: **3 / 5**. Leave-one-season-out CV mean R^2: **-0.00046** (see Section 4.3 for why a near-zero per-pitch R^2 here doesn't by itself decide the verdict - the same was true for this site's other models at the per-pitch level).

2. What Was Requested, What's Actually Here

The original spec: a gradient-boosted model fed ONLY characteristics of a pitcher's delivery - windup, timing, release - independent of Stuff+ and arsenal, to find which combination of delivery traits predicts good outcomes. The full requested feature list: height, weight, arm angle, extension, length of delivery, ball concealment, timing variance (holding the ball longer/shorter), quick pitches/leg hangs/timing disruptions, and release point.

Real investigation (same session) found three tiers:

- **Real and used here:** height, weight (pulled fresh from MLB Stats API's bio endpoint - not in Statcast at all), arm_angle, release_extension, release point (all Statcast fields), plus each pitcher's own season-long VARIABILITY in arm angle/extension/release point - the closest available proxy for “variation in the delivery.”
- **Real, but no accessible data source exists at all:** delivery length (windup-to-release duration) and ball concealment. Both would require frame-by-frame motion-capture data that MLB's Hawk-Eye system captures internally but does not publicly release. Not attempted - no honest proxy exists.
- **Real, partially prototyped, not yet solved:** leg lift timing, time held in the set position, and timing variance/quick pitches. A real pipeline (download broadcast video $\rightarrow$ pose-estimation with MediaPipe $\rightarrow$ derive joint-position timing) was built and PARTS of it work - video download and pose estimation both function on real MLB broadcast clips - but reliably telling which detected person in frame IS the pitcher (vs. catcher/umpire/batter) is unsolved as of this version. This is under separate, ongoing investigation, not included in this model.

So this v1 is explicitly a proof-of-concept on the reduced, honestly-available feature set - not the full model originally envisioned.

3. The Model

Same architecture as this site's Stuff+ v2 for consistency - LightGBM, Huber loss (delta_run_exp's heavy home-run-driven tail), season as a categorical feature to absorb era effects, leave-one-season-out cross-validation across 2021-2025 (2026 partial, train-only). The only real difference: no pitcher identity, no pitch type, no velocity, no movement, no plate location - ONLY the body/release features listed above, so the model can't quietly re-derive Stuff+ under a new name.

Target: **delta_run_exp (linear-weights run value per pitch, same as Stuff+ v2)**.

4. Validation

The same rigor Tunneling+ needed to prove it wasn't noise - this model's premise is more speculative than Stuff+'s (a much smaller, less obviously-predictive feature set), so it doesn't get a pass on validation.

4.1 Split-half reliability

Each pitcher's pitches were split into two random, independent halves; the score was computed separately on each half and correlated. A metric dominated by noise would show a correlation near zero.

A real methodological bug was caught and fixed while building this check. The first version reused the already-computed, full-season arm_angle_std/release_extension_std/release_consistency values for both halves, instead of recomputing them independently within each half - since those three features are season-long aggregates, both halves saw literally identical values for them, which trivially inflated the correlation to 0.999 regardless of whether the model had any real signal. The leaderboard from that same run was also contaminated with position players who'd thrown 3-58 mop-up-inning pitches (not real pitchers with a real delivery profile), since worked examples weren't filtered to the same qualifying bar the outcome-correlation check already used. Both are fixed in the numbers below: per-half features are now recomputed from only that half's own pitches, and the leaderboard uses the same 300-pitch minimum as everywhere else.

Cutoff	n pitchers	Correlation
min 50 pitches	721	0.877
min 100 pitches	630	0.841
min 150 pitches	566	0.856

4.2 Does it predict real outcomes?

deception_plus was correlated against each pitcher's real 2025 outcomes (n = 574 pitchers with 300+ pitches). Correlation with this project's own Stuff+ v2 model: 0.544. Each outcome's correlation was then recomputed AFTER removing whatever Stuff+ already explains (linear residualization) - a meaningfully positive partial r means the delivery features add real signal Stuff+ doesn't already capture.

2025 outcome	Raw r	Partial r (Stuff+ removed)
whiff_pct	+0.292	+0.038
chase_pct	+0.300	+0.160
swstr_pct	+0.362	+0.097
k_minus_bb_pct	+0.505	+0.307
xwoba	-0.567	-0.386

Correlation with overall Stuff+ itself (0.544) is notably higher than this site's other “+” models saw against Stuff+ - expected, and not a coincidence: release_extension is a feature in BOTH models (Stuff+ v2 uses it directly; it also feeds effective_speed there), so some real overlap between the two scores is baked in by construction, not just correlated behavior. The partial-correlation column is what actually matters here: whiff% and swinging-strike% drop to near zero once Stuff+ is removed - for those, this model is mostly just re-deriving Stuff+ through the shared extension feature. Chase%, K-BB%, and xwOBA-against retain a meaningfully positive partial correlation - real, independent signal beyond Stuff+ for those three specifically, even though it isn't for the pure swing-and-miss outcomes.

4.3 A note on the near-zero per-pitch R^2

CV mean R^2 = -0.00046 - alarmingly low-looking on its own, but this is not unique to this model: a single pitch's outcome is dominated by noise no delivery model can see (contact luck, defense, sequencing). Stuff+ v2 and Tunneling+ both faced the same issue at the per-pitch level and were validated instead via aggregate reliability and real outcome correlation (Sections 4.1-4.2) - the same standard applied here. The difference is those models' aggregate validation came back strong; Section 1 states plainly whether this one's does too.

4.4 Feature importance (SHAP)

Feature	Mean |SHAP|
release_consistency	0.0008
release_extension	0.0008
release_pos_z	0.0005
release_extension_std	0.0004
arm_angle	0.0004
arm_angle_std	0.0004
weight_lb	0.0004
release_pos_x_as	0.0003
height_in	0.0003
season	0.0003

4.5 Worked examples

A soft, secondary sanity check - real pitcher names at the extremes. Illustrative only, not independent statistical evidence.

Highest deception_plus:

Pitcher	Deception+	Pitches
Tyler Rogers	133	964
Trevor Megill	127	820
Yoshinobu Yamamoto	124	3315
Kenley Jansen	122	867
Ronny Henriquez	120	1157

Lowest deception_plus:

Pitcher	Deception+	Pitches
Noah Murdock	59	392
Chad Green	75	710
Tayler Scott	79	536
Chris Roycroft	79	394
Angel Chivilli	80	1022

4.6 Cross-validation detail

Held-out season	R^2	MAE	Pitches
2021	-0.00044	0.1237	697,920
2022	-0.00046	0.1176	715,489
2023	-0.00045	0.1224	722,954
2024	-0.00056	0.1195	718,990
2025	-0.00040	0.1201	722,891

5. Known Limitations and Open Questions

- **Thin feature set.** Five real per-pitch/aggregate features stand in for what was meant to be a much richer delivery-timing profile - see Section 2.
- **Height/weight are current-snapshot, not season-specific.** Same limitation this site already accepts for team_id elsewhere - a pitcher's 2021 listed weight may differ slightly from what was pulled.
- **No shrinkage applied** to thin-sample pitchers, unlike Stuff+ v2.
- **Single season report.** Unlike Stuff+/Tunneling+, this proof-of-concept only validates against 2025 - no season dropdown yet, no year-over-year predictive-validity check.
- **The video-based delivery-timing pipeline is a live, separate investigation** - not blocked forever, just unsolved as of this version. See Section 2.

6. Suggested Next Steps

- Solve reliable pitcher-vs-other-players isolation in broadcast video (uniform color + position heuristics, or a properly-isolated player-detection model), then add leg-lift/set-time/timing-variance features for real.
- Multi-season report + season dropdown, matching Stuff+/Tunneling+.
- Season-specific height/weight if a historical bio data source can be found.
- Empirical-Bayes shrinkage for thin-sample pitchers.